

9.4 SHORT CIRCUIT OF A LOADED SYNCHRONOUS MACHINE

In the previous article on the short circuit of a synchronous machine, it was assumed that the machine was operating at no load prior to the occurrence of short circuit. The analysis of short circuit on a loaded synchronous machine is complicated and is beyond the scope of this book. We shall, however, present here the methods of computing short circuit current when short circuit occurs under loaded conditions.

Figure 9.10 shows the circuit model of a synchronous generator operating under steady conditions supplying a load current I^o to the bus at a terminal voltage of V^o . E_g is the induced emf under loaded condition and X_d is the direct axis synchronous reactance of the machine. When short circuit occurs at the terminals of this machine, the circuit model to be used for computing short circuit current is given in Fig. 9.11a for subtransient current, and in Fig. 9.11b for transient current. The induced emfs to be used in these models are given by

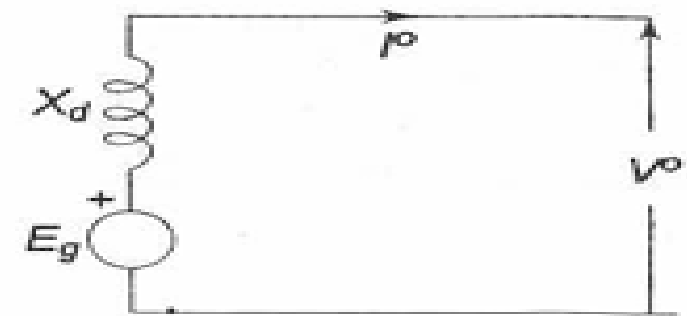


Fig. 9.10 Circuit model of a loaded machine